

Assessing the Generalizability and Robustness of Deep-Learning Dose Prediction in Head-and-Neck Radiotherapy to Clinically Realistic CT Perturbations

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Abstract

Purpose: Deep-learning dose prediction models are increasingly deployed across institutions, where they encounter computed tomography (CT) images acquired on different scanners, with different reconstruction kernels, dose levels, and artifact content. In contrast to worst-case adversarial perturbations, these represent benign, physically plausible sources of input variability. This study quantifies the robustness of a head-and-neck dose prediction model to five families of clinically realistic CT perturbations and identifies the image-quality factors that materially affect predicted dose.

Methods: A 3D U-Net dose predictor with squeeze-and-excitation attention (baseline dose score 3.73 Gy, DVH score 2.53 Gy on the OpenKBP benchmark) was evaluated on 40 held-out test patients under 26 CT input conditions: an unperturbed baseline and five perturbation families, each applied at five severity levels. The families model acquisition noise, Hounsfield-unit (HU) calibration shift, low-frequency bias field, spatial-resolution loss, and dental streak artifact, with severity ranges chosen to meet or exceed American College of Radiology (ACR) CT-simulation quality-assurance limits. Perturbations were applied to the model input CT only; structure sets were held fixed, and each perturbed prediction was compared with the same patient’s unperturbed prediction. A perturbation level was defined as clinically visible when the cohort-mean change in any of 23 dose-volume histogram (DVH) criteria exceeded 1.0 Gy.

Results: Four of the five families produced no clinically visible change at any tested severity, including severities well beyond clinical plausibility. Spatial-resolution loss was the only family to cross the threshold: the larynx near-maximum dose ($D_{0.1cc}$) exceeded 1.0 Gy at a 2.0/1.0-voxel anisotropic blur, a level within realistic cross-scanner variation in slice thickness and reconstruction kernel, and reached 2.84 Gy (an 18.2% DVH-score degradation) at the highest severity. Bone-weighted HU calibration shift produced a measurable but sub-threshold effect only at an implausible 1000 HU offset (0.61 Gy). Acquisition noise, bias field, and dental streaks never exceeded 0.34 Gy on any criterion.

Conclusions: The model is robust to intensity-based CT variability but systematically sensitive to spatial-resolution degradation, which becomes clinically visible within the range of realistic inter-scanner variation. Multi-institution quality assurance for learned dose prediction should prioritize CT spatial-resolution and reconstruction-kernel consistency over intensity calibration, and dose-based rather than image-quality metrics should be used to assess the suitability of CBCT and synthetic CT inputs.

1 Introduction

Deep-learning dose prediction is increasingly used to support automated, knowledge-based, and adaptive radiotherapy treatment planning [1]. A model trained at one institution must generalize to CT images acquired under different conditions: scanner hardware, reconstruction kernel, tube current, slice thickness, and patient-specific artifacts all vary across and within clinics. Evaluation of these models, however, is typically limited to prediction accuracy on a held-out test set drawn from the same distribution as the training data. Consequently, model robustness to realistic input variability remains poorly characterized, and accuracy-based assessments may overlook failure modes that affect clinically relevant outputs.

Prior work, including our own, has probed the sensitivity of dose prediction networks with adversarial perturbations such as the fast gradient sign method (FGSM) and projected gradient descent (PGD) [4]. Such attacks establish worst-case bounds but answer a different question from the one facing a clinic that deploys a trained model on a new scanner: adversarial perturbations are optimized against the model, whereas real-world image variability is benign, structured, and physically constrained. A model may be fragile under adversarial attack yet perfectly stable under any input change that can actually occur in practice, or vice versa.

This study addresses the deployment question directly. We construct five families of physically motivated CT perturbations, each a controlled model of one real-world source of image variability: quantum acquisition noise, scanner HU calibration drift, low-frequency intensity inhomogeneity, loss of spatial resolution, and dental metal streak artifact. Each family is applied over a five-level severity sweep spanning routine clinical variation, ACR CT-simulation quality-assurance tolerances, and deliberately supra-clinical extremes. We then determine, per family, the severity at which the perturbation first produces a clinically significant change in predicted dose. The result is a ranking of CT image-quality factors by their relevance to the safe multi-institution deployment of a learned dose predictor.

2 Methods

2.1 Dataset

All experiments use the OpenKBP head-and-neck cohort [1], a public benchmark of 340 oropharyngeal cancer patients planned for intensity-modulated radiotherapy with prescriptions of 70, 63, and 56 Gy delivered in 35 fractions. Each patient comprises a CT volume and binary masks for up to ten regions of interest: seven organs at risk (brainstem, spinal cord, right and left parotid glands, esophagus, larynx, mandible) and three planning target volumes (PTV56, PTV63, PTV70). All volumes are provided on a $128 \times 128 \times 128$ voxel grid with approximately $3.9 \times 3.9 \times 2.5$ mm³ voxels. We use the standard partition of 200 training and 40 held-out test patients; all robustness results are reported on the 40 test patients. CT intensities are stored as non-negative values offset such that air is near 0 and water near 1024 (true HU plus 1024), clipped to [0, 4095], and normalized to [0, 1] for network input. Dose is normalized by the 70 Gy prescription.

2.2 Dose prediction model

The predictor is a 3D U-Net [2] with six encoder stages (64 to 512 filters) and five decoder stages with skip connections. Instance normalization is used throughout, residual connections are included within encoder and decoder blocks, and squeeze-and-excitation channel attention [3] is applied at the deepest levels. The network receives the CT volume and the ten structure masks and outputs

a voxel-wise 3D dose distribution. It was trained for 100 epochs with a masked mean absolute error loss restricted to the feasible dose region, a $4\times$ loss weighting on PTV voxels, mixed-precision arithmetic, and flip and intensity-scaling augmentation. On the OpenKBP test set the model achieves a dose score (voxel-wise mean absolute dose error within the feasible region) of 3.73 Gy and a DVH score (mean absolute error over the standard OpenKBP DVH criteria) of 2.53 Gy, compared with 2.43 and 1.48 Gy for the competition-winning ensemble. The model is used exactly as trained; no fine-tuning, test-time adaptation, or perturbation-aware augmentation is applied in this study.

2.3 Perturbation families

Each family perturbs the CT volume only, inside the patient body mask, with all outputs clipped back to the valid intensity range $[1, 4095]$. Structure masks are never modified. Randomness is controlled by a deterministic seed derived from the (family, level, patient) triple, making the entire perturbed dataset exactly reproducible. Severity parameters for all levels are listed in Table 1.

P1: Signal-dependent acquisition noise. Zero-mean Gaussian noise with a per-voxel standard deviation that is larger in bone than in soft tissue, modeling the heteroscedastic quantum noise of CT detectors:

$$\tilde{x}(v) = x(v) + \varepsilon(v), \quad \varepsilon(v) \sim \mathcal{N}(0, \sigma(v)^2), \quad \sigma(v) = \begin{cases} \sigma_{\text{bone}} & x(v) > 1500 \\ \sigma_{\text{soft}} & \text{otherwise,} \end{cases} \quad (1)$$

with $(\sigma_{\text{soft}}, \sigma_{\text{bone}})$ swept from $(8, 12)$ to $(100, 160)$ HU. Typical single-scanner noise is approximately 10 to 50 HU, so the upper levels substantially exceed clinical reality.

P2: Bone-weighted HU calibration shift. A systematic intensity offset with random sign, interpolated smoothly between a soft-tissue value and a larger bone value through a sigmoid gate centered at 1500 stored HU with width 200:

$$\tilde{x}(v) = x(v) + s \left[\Delta_{\text{soft}} + (\Delta_{\text{bone}} - \Delta_{\text{soft}}) \sigma\left(\frac{x(v) - 1500}{200}\right) \right], \quad s \in \{-1, +1\}, \quad (2)$$

modeling calibration drift and cross-scanner HU differences, which are typically 10 to 50 HU in soft tissue. The sweep extends to a deliberately implausible $(100, 1000)$ HU offset.

P3: Low-frequency bias field. A smooth multiplicative-style shading field added within the body, constructed as the sum of three separable 3D cosine harmonics with random frequencies (0.5 to 2.5 cycles per field of view), phases, and weights, normalized to unit peak and scaled to an amplitude swept from 10 to 200 HU. This models scatter- or calibration-induced spatial nonuniformity, which is realistically tens of HU.

P4: Spatial-resolution loss. Anisotropic Gaussian blur with through-slice standard deviation σ_z twice the in-plane σ_{xy} , modeling thicker slice acquisition and smoother reconstruction kernels:

$$\tilde{x} = G_{(\sigma_z, \sigma_{xy}, \sigma_{xy})} * x, \quad (3)$$

with (σ_z, σ_{xy}) swept from $(0.5, 0.25)$ to $(4.0, 2.0)$ voxels (levels L0 to L4). Realistic cross-scanner variation in slice thickness and kernel corresponds to roughly 1 to 3 voxels of effective blur on this grid.

P5: Dental streak artifact. A radial pattern of alternating bright and dark streaks anchored at the mandible centroid, with Gaussian angular profiles, radial decay (40-voxel length constant), Gaussian through-slice decay (3-slice standard deviation), and a saturated metal spot (3-voxel radius) at the centroid. Amplitude and streak count are swept from (150 HU, 8 streaks) to (1200 HU, 24 streaks). Dental artifacts of this character are common in head-and-neck CT.

2.4 Evaluation protocol

The 26 conditions (baseline plus five families at five levels each) define 26 versions of the 40-patient test set, 1040 model inferences in total. For each condition the model re-predicts dose from the perturbed CT with the original structure masks. Impact is quantified against each patient’s own unperturbed prediction, isolating the causal effect of the image perturbation from baseline model error. Two metric groups are computed:

- **Voxel-wise:** mean absolute dose difference (MAE, Gy) within the feasible dose region, aggregated as the OpenKBP dose score; and the fraction of patients whose MAE change exceeds 0.5 Gy.
- **DVH criteria:** the 23 standard OpenKBP criteria, namely $D_{0.1cc}$ and mean dose for each organ at risk, and D_{99} , D_{95} , and D_1 for each target, aggregated as the OpenKBP DVH score and also examined per criterion.

Clinical visibility threshold. A severity level is defined as *clinically visible* when the cohort-mean absolute change in *any single* DVH criterion, relative to each patient’s unperturbed prediction, exceeds 1.0 Gy. The per-criterion definition is deliberately more sensitive than a threshold on the aggregate DVH score, which averages over 23 criteria and can mask a large error confined to one structure. The fraction of patients with MAE change above 0.5 Gy is reported as a corroborating metric but is not part of the gate.

3 Results

3.1 Aggregate degradation

Table 2 reports the dose score and DVH score for all 26 conditions. Four families are essentially flat across their full sweeps: the largest aggregate DVH-score change is +1.43% for acquisition noise at (100, 160) HU, +0.42% for the bias field at 200 HU, and +0.87% for dental streaks at 1200 HU, all at severities well beyond clinical ranges. The HU calibration shift is flat through L3 and becomes measurable only at the implausible extremes, reaching +11.2% DVH score at a 1000 HU bone offset. Spatial-resolution loss is the clear outlier: degradation grows approximately linearly with blur, from +1.5% DVH score at near-native resolution (L0) to +18.2% at L4, with the dose score rising from 3.73 to 4.12 Gy.

3.2 Per-criterion threshold analysis

The clinical-visibility analysis examines the worst single DVH criterion per condition (Table 3). Spatial-resolution loss is the only family that crosses the 1.0 Gy threshold anywhere in the tested range. The sentinel criterion is the larynx near-maximum dose ($D_{0.1cc}$), which shifts by 0.22 Gy at L0, 0.59 Gy at L1, 1.15 Gy at L2 (the first crossing), 1.68 Gy at L3, and 2.84 Gy at L4. At L4 a second criterion also exceeds 1 Gy. The corroborating voxel-wise metric follows the same pattern:

Table 1: Severity parameters for the five perturbation families. P4 uses levels L0 to L4; all other families use L1 to L5.

Family	Parameter (units)	Level				
		1	2	3	4	5
P1 noise	$\sigma_{\text{soft}}/\sigma_{\text{bone}}$ (HU)	8/12	15/25	30/50	60/100	100/160
P2 HU shift	$\Delta_{\text{soft}}/\Delta_{\text{bone}}$ (HU)	5/50	10/100	25/250	50/500	100/1000
P3 bias field	amplitude (HU)	10	20	50	100	200
P4 resolution	σ_z/σ_{xy} (voxels)	0.5/0.25	1.0/0.5	2.0/1.0	3.0/1.5	4.0/2.0
P5 dental	amplitude (HU) / streaks	150/8	300/12	500/16	800/20	1200/24

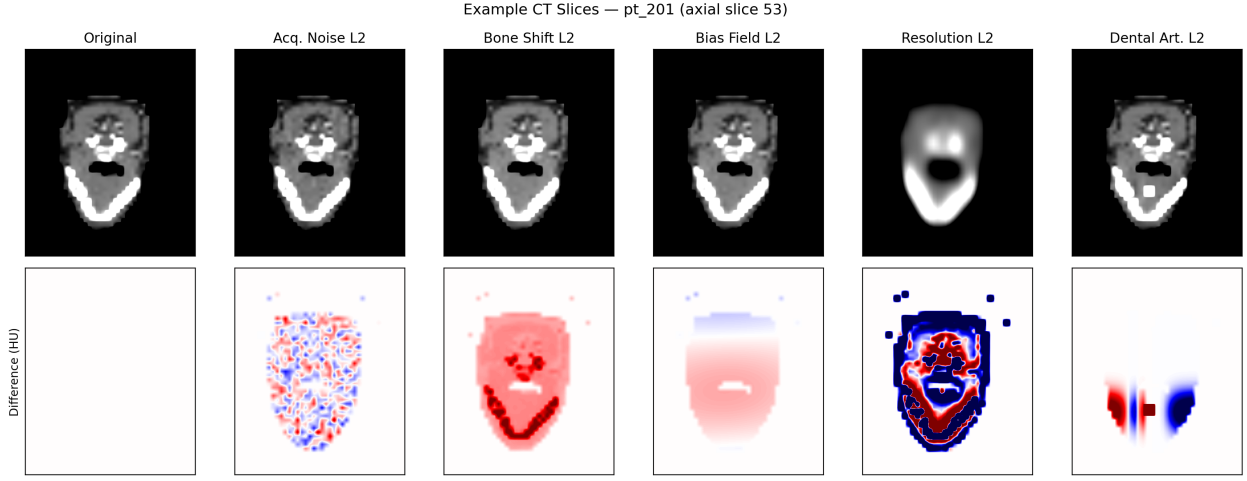


Figure 1: Axial CT slice (patient pt_201) unperturbed and under each perturbation family at an intermediate severity (top), with HU difference maps (bottom). Intensity families preserve anatomical boundaries; resolution loss erodes them.

the fraction of patients with an MAE change above 0.5 Gy grows from 0.00 at L0–L1 to 0.05, 0.12, and 0.23 at L2, L3, and L4.

No other family reaches the threshold on any criterion at any severity. The HU calibration shift peaks at 0.61 Gy (PTV56 D_1) at the 1000 HU offset; acquisition noise peaks at 0.14 Gy (spinal cord $D_{0.1cc}$); the bias field at 0.09 Gy (right parotid mean); and dental streaks at 0.33 Gy (PTV70 D_1).

3.3 Spatial and structural error patterns

Example perturbed CT slices and the corresponding dose-difference maps are shown in Figures 1 and 3. The four intensity families alter voxel values while leaving anatomical boundaries intact, and their dose-difference maps are near zero and spatially unstructured. Resolution loss visibly erodes tissue boundaries, and its dose errors are spatially structured, concentrating at structure boundaries and in the dose-gradient regions between targets and adjacent organs at risk. Per-structure analysis at the highest severity shows the largest relative error inflation in geometrically complex, boundary-dependent structures: the larynx, the parotid glands, and the mandible. The severity progression for resolution loss (Figure 2) shows the approximately linear growth of the worst-criterion shift with blur and the crossing of the 1.0 Gy line between L1 and L2.

Table 2: Dose score (voxel-wise MAE, Gy) and DVH score (Gy) for all 26 conditions, 40 test patients. Deltas are relative to the unperturbed baseline.

Family	Level	Dose score (Gy)	DVH score (Gy)	Δ DVH (Gy)	Δ DVH (%)
Baseline	–	3.731	2.535	–	–
P1 noise	L1	3.731	2.535	0.000	0.01
	L2	3.733	2.538	0.003	0.13
	L3	3.737	2.543	0.009	0.34
	L4	3.742	2.559	0.024	0.96
	L5	3.742	2.571	0.036	1.43
P2 HU shift	L1	3.730	2.531	−0.004	−0.15
	L2	3.740	2.544	0.010	0.38
	L3	3.751	2.519	−0.016	−0.61
	L4	3.793	2.572	0.037	1.46
	L5	3.916	2.818	0.284	11.19
P3 bias field	L1	3.732	2.535	0.000	0.01
	L2	3.730	2.534	−0.001	−0.04
	L3	3.736	2.534	−0.000	−0.01
	L4	3.733	2.524	−0.011	−0.44
	L5	3.759	2.545	0.011	0.42
P4 resolution	L0	3.739	2.573	0.038	1.50
	L1	3.780	2.636	0.101	4.00
	L2	3.885	2.729	0.194	7.67
	L3	3.982	2.824	0.289	11.40
	L4	4.124	2.995	0.460	18.15
P5 dental	L1	3.729	2.541	0.007	0.26
	L2	3.730	2.545	0.011	0.42
	L3	3.730	2.552	0.017	0.67
	L4	3.731	2.557	0.022	0.87
	L5	3.732	2.555	0.021	0.82

4 Discussion

The central finding is a sharp asymmetry between intensity-based and geometry-based input variability. The model tolerates intensity perturbations at severities far beyond clinical plausibility: noise at $2\times$ to $3\times$ typical scanner levels, calibration offsets an order of magnitude above realistic drift, shading fields of 200 HU, and dense high-amplitude streak artifacts all leave every DVH criterion within 1 Gy of the unperturbed prediction. A plausible mechanism is that these perturbations preserve the structural boundaries that the network, which also receives the structure masks as input, uses to localize dose gradients; zero-mean or slowly varying intensity changes are largely averaged out by the convolutional feature hierarchy.

Spatial-resolution loss breaks exactly this invariance. Blur removes the high-frequency boundary information on which sharp dose gradients are conditioned, and the resulting errors are spatially structured, appearing first at the interfaces of small, geometrically complex structures. The sentinel criterion, larynx near-maximum dose, is a near-maximum metric in a small organ embedded between the pharyngeal target volumes, which is precisely where boundary localization matters most. Importantly, the first threshold crossing occurs at a 2.0/1.0-voxel blur, corresponding to roughly 5 mm through-slice and 4 mm in-plane full-width-at-half-maximum smoothing on this grid, which

Table 3: Worst single-criterion DVH shift (cohort mean, Gy) per family, the sentinel criterion, the number of criteria exceeding 1.0 Gy, and the fraction of patients whose voxel-wise MAE rises by more than 0.5 Gy.

Condition	Max shift (Gy)	Sentinel criterion	# > 1 Gy	Frac. MAE > 0.5 Gy
P1 noise, L5 (worst)	0.14	Spinal cord $D_{0.1cc}$	0	0.00
P2 HU shift, L4	0.41	Larynx $D_{0.1cc}$	0	0.03
P2 HU shift, L5	0.61	PTV56 D_1	0	0.17
P3 bias field (worst)	0.09	Right parotid mean	0	0.00
P4 resolution, L0	0.22	Larynx $D_{0.1cc}$	0	0.00
P4 resolution, L1	0.59	Larynx $D_{0.1cc}$	0	0.00
P4 resolution, L2	1.15	Larynx $D_{0.1cc}$	1	0.05
P4 resolution, L3	1.68	Larynx $D_{0.1cc}$	1	0.12
P4 resolution, L4	2.84	Larynx $D_{0.1cc}$	2	0.23
P5 dental (worst)	0.33	PTV70 D_1	0	0.00

is within the realistic range of cross-scanner slice thickness and reconstruction-kernel differences. Resolution mismatch is therefore not a tail risk but an expected condition of multi-institution deployment.

These results complement adversarial robustness analyses. Our earlier adversarial evaluation of the same class of models showed that optimized perturbations far smaller than any of the levels used here can produce large dosimetric errors. The present study shows that among perturbations that actually occur in practice, only resolution loss carries clinically relevant risk for this model. The two analyses answer different questions: adversarial attacks bound the worst case, whereas realistic perturbation sweeps rank the operational risks. For deployment decisions the realistic ranking is the actionable one, and it points at CT spatial-resolution consistency, not intensity calibration, as the property to control.

Three practical implications follow. First, commissioning and ongoing quality assurance for learned dose prediction should explicitly include a resolution-mismatch test, for example by predicting on a resampled or kernel-smoothed copy of a commissioning CT and verifying DVH stability. Second, image-quality metrics tuned to intensity fidelity (noise, uniformity, HU accuracy) are insufficient proxies for model safety; dose-based end-to-end checks are required, particularly when considering CBCT or synthetic CT inputs whose principal defects include resolution and boundary degradation. Third, the approximately linear severity response of the resolution family suggests that training-time blur augmentation may flatten the curve, which is directly testable.

4.1 Limitations

This is a single-model, single-anatomy study on a public benchmark; the threshold levels reported are specific to this network, training recipe, and voxel grid, and the generality of the resolution failure mode across architectures remains to be established (see Section 5). Structure masks were held fixed under perturbation, isolating the image pathway; in clinical reality, degraded images would also degrade contouring, likely compounding the effect. The perturbation families are physically motivated simulations rather than paired rescans, and the dental artifact model is a geometric approximation of beam-hardening physics. The 1.0 Gy per-criterion threshold, while grounded in the granularity of clinical plan evaluation, is a choice; the full severity curves are reported so that any alternative threshold can be applied. Finally, gamma analysis was not computed; the evaluation is DVH- and voxel-MAE-based.

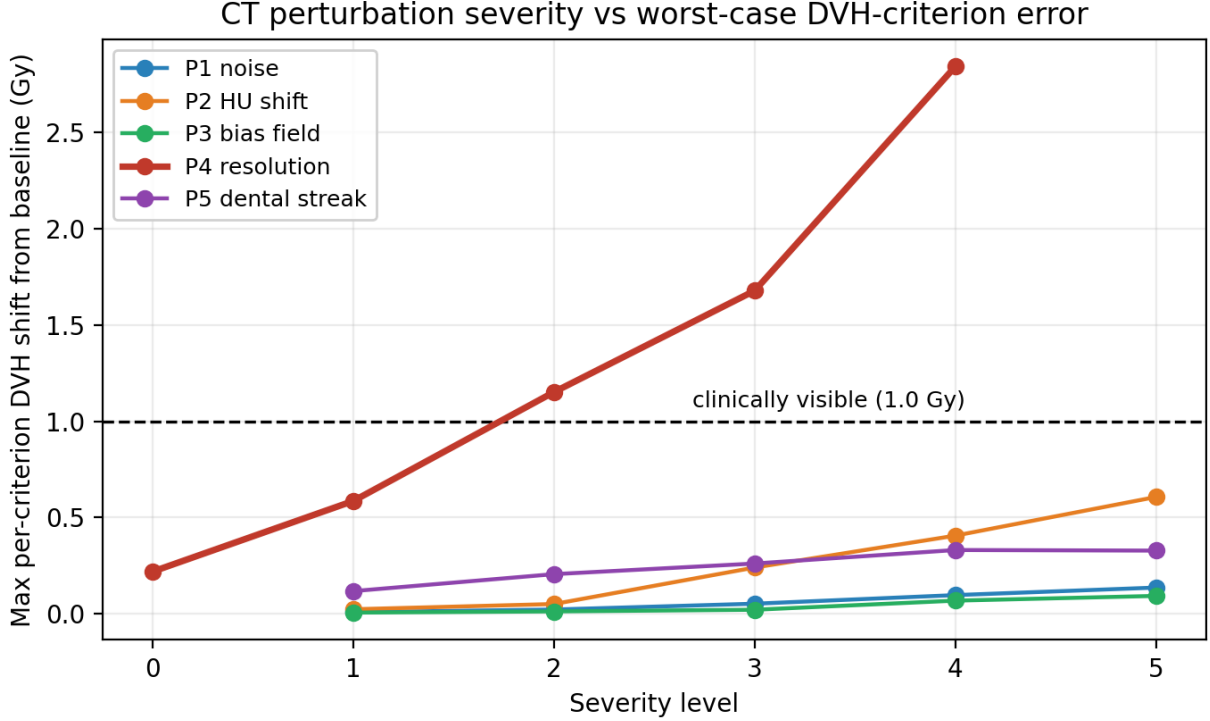


Figure 2: Worst single-criterion DVH shift (cohort mean, Gy) versus severity level for the five families. The dashed line marks the 1.0 Gy clinical visibility threshold. Only spatial-resolution loss (P4) crosses it, between L1 and L2.

5 Future work

Three extensions are planned. First, a multi-architecture comparison deploying four to five distinct dose prediction models (a plain 3D U-Net, the present SE U-Net, an attention or transformer-based variant such as Swin UNETR, and a cascade or ensemble representative of the OpenKBP state-of-the-art) under the identical 26-condition battery, to determine whether the resolution sensitivity and its L2 crossing point are architecture-specific or a property of the task. The perturbation and evaluation pipeline is model-agnostic (CSV volumes in, CSV doses out), so each additional model requires only 1040 inference passes. Second, CBCT-characteristic degradations (scatter and cupping, ring artifacts, and limited field-of-view truncation) to establish commissioning tolerances for online adaptive radiotherapy. Third, distribution-free and certified per-patient DVH prediction intervals under bounded CT perturbation, and training-time resolution augmentation as a mitigation.

6 Conclusions

A head-and-neck deep-learning dose prediction model tolerates clinically realistic intensity-based CT perturbations, including acquisition noise, calibration drift, bias fields, and dental streak artifacts, at severities meeting or exceeding ACR quality-assurance limits. It is systematically sensitive to spatial-resolution degradation, which becomes clinically visible (larynx $D_{0.1cc}$ shift above 1 Gy) within the realistic range of inter-scanner slice-thickness and reconstruction-kernel variation and

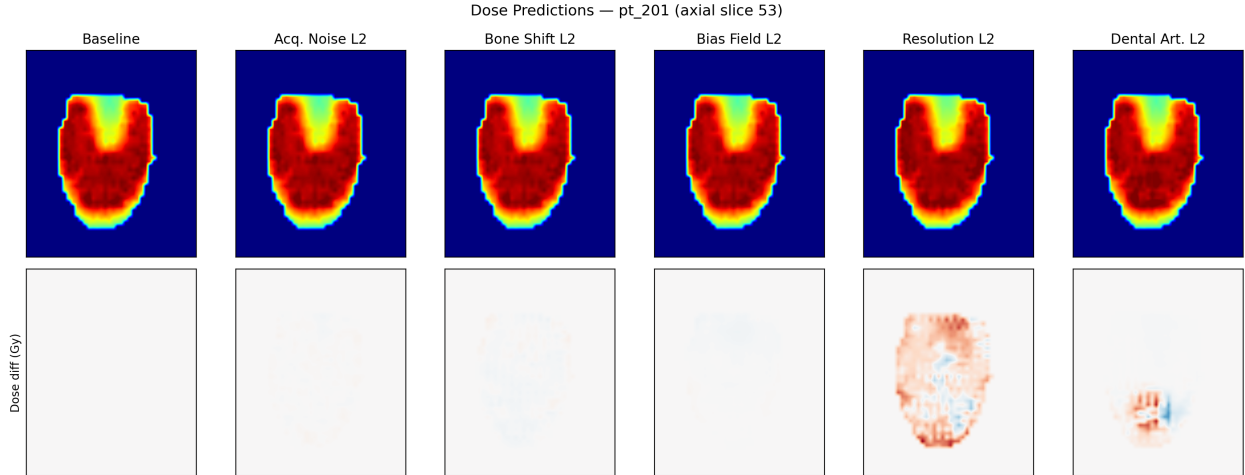


Figure 3: Predicted dose (top) and difference from the unperturbed prediction (bottom) for each family at an intermediate severity. Resolution loss produces spatially structured errors at anatomical boundaries; the other families are near zero.

grows approximately linearly with blur. Multi-institution quality assurance for learned dose prediction should prioritize CT spatial-resolution and reconstruction-kernel consistency, and dose-based rather than image-quality metrics should govern the acceptance of alternative image sources.

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